ActInf Textbook Group ~ Cohort 4 ~ Meeting 17 (Chapter 8 part 1)

Source: YouTube **Playlist**: Active Inference Textbook Group ~ Cohort 4
Duration: 54:44 | **Uploaded**: Unknown **Transcript method**: auto_caption

---

all right [Music] welcome it's October 17th 2023 and we're in our first discussion of chapter 8 in cohort 4 so to begin with what are just any thoughts or memories that you fellows have about chapter 8 just any any piece that you remember or anything about eight um I was curious about the part where they talk about how movement and attenuation like sensory attenuation that was interesting to me where they I hope I remember calling this correctly but it was like you didn't want to perfectly predict it at a certain point for like to perceive movement which was interesting um I don't know if I perfectly understand that but uh I kind of see an intuitive aspect to that I guess oh yeah that's a very interesting section Andrew anything that that you um think about yeah yeah um I guess I'd second an interest in sensory attenuation as well because I I guess um we could get to that but like box 8.1 is kind of the key bit of interest there the relationship between Precision attention and sensory attenuation so they seem to more or less equate precision and attention um I guess the idea is um you have some kind of belief that is you know I am not moving and you need you need to downplay or decrease the Precision of that belief such that you can actually move in the first place otherwise if you have very high Precision I am not moving um that kind of stays in place and so you never actually get started on moving um yeah know I thought that was really interesting and it comes up a little bit more later in the chapter on how that kind of plays out and the movement of Limbs and such so I thought that was pretty pretty fascinating yeah totally and like where else does that apply like can we understand the example they give with a motor Behavior connected to the Octan fonty and then what other analogous um sensory attenuation cases might be happening let's just see if it's been explored in a question okay we'll definitely look at this question about precision and then here's the Box 8.1 okay no answer what what what's interesting about it um a or what what would you want to know about it or what do I mean um I guess like I was just it was just something where um when I read it it was it made uh sense but I'm still trying to let me see if I can find it it made sense it was just um interesting that we aren't like was it were they saying that the perception of movement itself like in order to perceive movement it was like we were decreasing how um well we were predicting uh those States yeah in order to generate Movement we must be able to ignore the sensory consequences which is to say dis attend to it or attenuate to it yeah of that move movement to form the initially false belief that I am moving okay okay yeah that's like just I don't know something about that's very interesting it's a so and then afterwards are you we are trying to make up for that by uh trying to predict those uh or get those appropriate like inferences right um it's not you're not you're you're going to start attending to it again or is it you just decrease how much attention you're putting on to those States at first and then after you're yeah make certainly it's an oscillatory pattern so one one analogy or another case or sensory tenation is really important is the ey Cade so when the Eyes Are Fixed the Precision is is high so that changes in the visual scene when the Eyes Are Fixed can be informative and then during head movement Andor eye movement you kind of have like a a hyper prior that pixel changes or or um are uninformative because they're they're not going to be very useful there's You're Expecting High volatility on when there's movement so during that there like a sensory attenuation during the seccade and then as soon as the Cade ends it's just like re-engages attenion okay so it's kind of like when when the um when the signal to contract the bicep occurs there's also a descending relaxation of the tricep in in the kind of healthy setting so it's not just like pull on one side and pull on the other side it's like a joint contraction and relaxation signal that that's happening with two opponent muscles whereas the icade attenuation example is more like opponent processing with a motor and an attention variable but they they alternate okay so we have the isoc case with visual perception getting out of a chair well and there probably are other differences here like you could argue whether the iade cons it's kind of like it's considering the counterfactual um just like the getting out of the chair is I am not in a chair is the counterfactual the iade counterfactual is like I am looking over there but you're not and so the the um distri the belief distribution um ultimately you want to shift your location in that space But if the um current Precision around the current state estimate doesn't have significant Mass covering that counterfactual then it it can't be switched over to so sensory attenuation relaxes the attention expands the variance reduces the Precision makes it so it's like instead of like 99% sure that it's this way go well it's 80% sure that it's this way 20% that it's that way that opens the door to the counterfactual being realized through action just to kind of restated a bunch of ways because I think if it is the case that that's how um attention and attenuation work then probably it's going to be prevalent in like every sensory motor system because the alternative would be constant Precision yeah and then would so constant Precision would make it would actually make you less likely to um be able to process movement because there's a lot of uh it's almost like you're be able like less likely to get new information is that kind of a way of thinking about it is that why we're kind of relaxing the attention so we could be more flexible with sort of uh states that are changing very rapidly I think that's true so let's think about what would happen if you had constant Precision constant Precision to your visual field in iade or constant Precision to your proprioception for the getting out of the chair case well when your eyes were um moving just where would you fix your constant value if you fixed it at the level of that that your surprise should be at for being fixed then it'll be shocking when there's movement if you have it when it is if you um calibrate to when the pixels are moving a lot then you won't be able to glean as much information when they're not moving because you'll be dis attending or you could pick a lukewarm value and kind of blend those two be a little bit more surprised than you should be when it's surprising and then a little bit less informative when your gaze is fixed so obviously an approach to resolve this tension is to have oscillation or some kind of conditional attention so that you can tune it and do the best you can do in each moment instead of trying to like fix a value when we already know that there's a different optimal Precision for different um settings or like with through time within the same setting okay that makes sense it's kind of like a uh trying to think of a good sort of analogy but uh just like you're actually doing worse by it's kind of like you know you're grasp in like have you ever heard of that analogy where you're grasping like a seed you can't hold too hard or it'll like slip out but if you hold it too loose it'll also slip out so you need to like adjust yeah that's that's good connection that's also related to kind of optimal grasp like Optimal grasp on a pen is not the maximum compress Force for for a variety of reason you know you'd be exhausted you couldn't have um dextrous movement or something like that so like then that's related to like not over under fitting and that's kind of the VF imperative which is you want to fit a distribution given the family of variational distributions that you're choosing from gaussian or whatever it is you want to parameterize it so that if you were to make it any tighter you'd be overfitting and then if you made it any looser you would be having left information that you could have used so it's on some trade-off Frontier um which we can be along with a KL Divergence so it's like a it's you know it's in a good spot and it's on a good optimizable manifold that if the variational family is at all appropriate you're going to get to the optimal grasp now maybe it's like um trying to do optimal grasp on something that's too small or too large to be grasped that would be like if the distribution family was inappropriate then you'd You' come to the best grasp you could of you know the mountain but you there you it wouldn't be possible to actually do it so that kind of con actually like the the the model that doesn't under overfit still may have an adequacy of 1% given the data and all this other things MH and then and then one other connection there is like um the Colman filter or just generalized filtering so we're getting measurements from the thermometer Through Time one or more thermometers through time and then we're tracking the temperature distribution um or stock prices or something then you want to be tracking both the expected mean temperature and the variance like the volatility or just uncertainty the the sort of fundamental uncertainty or volatility and then also the measurement Associated so the measurement Associated volatility is more like a matrix the actual Dynamics are more like B but and so but to whatever extent it comes the noise or the variability comes from either the sensor measurements or the underlying dynamics of the system you you want to be fitting this curve Through Time such that new noisy data measurements are not like falling outside the curve because then that's more surprising than it should be then you should wide to the curve or um if all the data are falling like within the plus or minus one standard deviation on the curve then you could have done better by tightening the distribution son's work on the generalized filtering is really nice on this on the the generalized filtering kind of connecting into the Baye um basing optimal filtering in perception and then bringing in action this also kind of reminds me of uh exploration exploitation tradeoff and reinforcement learning and uh soft value functions where they actually instead of just trying to optimize a reward they try to also increase the entropy like an entropy value um so which seems kind of counterintuitive but it's actually performs really well it makes it very flexible and able to deal with protuberances specifically so like disturbances like a if you have a robot that's trying to work in an environment where things can start disturbing it and things like that having sort of that like trying to optimize that entropy value as well as the reward value makes it so it can actually adjust and uh deal with disturbances a lot better which kind of reminds me of like movement also it's kind of like disturbances to sort of your oh yeah yeah and uh there's two kinds of motor disturbances ones you have agency over and ones you don't have agency over which ones are more surprising the ones you don't have agency over like um you can um not a recommendation but you can slightly push the eye so that it's kind of like a non seccade induced visual shift of course that's surprising um or if uh you can't tickle yourself because the expected propri acceptive consequences of tickling yourself are not attended to because they're expected so they're confirmed um or yeah just how we expect H how far do we expect the stair to be if it's a little bit more or less then it's surprising so then just the expected consequences of action are normalized too I mean Sensations are perceived calibrated to what their expected states are this is kind of the whole predictive processing everything is is is like sensory attenuation is kind of a Cory of um a first principles corollary of this kind of action and perception as inference and then it just like um uh predictive coding in the visual system this is empirically found in different systems and then and then it'd be interesting like what other phenomena are like where do we see allostasis or attenuation in um in other systems like if you have um the bacteria and the gradient of sugar or like antibiotics or something it recalibrates to the value with like a kind of log sensitivity so that it can always have optimal grasp on making a decision about the gradient if the gradient were a thousand times longer than the bacteria it couldn't detect it if the gradient was a millionth of the wavelength of a bacteria it couldn't detect it but then there's some range that it can actually do the calibration that gives the most statistical resolution just like the aperture of a camera or the noise level of a sensor and so then when you're calibrating the microphone or the the whatever it is the two kind of two minimal things or the one minimal thing is just the mean that or that's like the Delta the dero Delta function is if you can only estimate one parameter about to calibrate the microphone in the room you would want to know the potentially the average or the mean of the median volume and then the second thing would be a variance parameter um those might be jointly estimated for certain one parameter families of distributions like Pon distribution where you only use one parameter to describe the family and then that is like where like the mean and the variance are equal is a one parameter estimation but in other case the kind of two that are helpful are the central tendency and then the variance it doesn't have to be gaussian but you kind of need those two because if you don't have a measure of this of a central tendency and then how wide a field of the central tendency then it's like you're kind of saying that it's a diverging situation so then it's not really identifiable good good stats pieces all of regular these are common topics and in other statistics settings and it's kind of cool because like they're often discussed in the signal processing inbound passive inference side common filter all the variational auto encoder and then also in in a kind of slightly siloed way in the control theory optimal control K optimal control way so then those two statistical problems can be understood as kind of like two complimentary special cases there's optimal signal processing when you have no agency or no action control and then there's optimal control in a completely known environment where there's no epistemic value which is like the figure from earlier with showing expected free energy and then when certain things are included or not what special cases results so sensory attenuation it's a useful tractable Motif that does that that uh addresses the the the kind of realistic statistical problems by modulating and learning attention another thing I was uh curious about was the generalized synchrony um where they're talking about the birds um doing the bird song and then the the other one would try to finish the other segments of the song it just kind of started going into uh something I'm like uh been interested in which is like communication and uh even like Linguistics and just like how that seems similar to communicating language and like when you talk to like a language model you're like people a lot of people say it's just predicting sort of the the best response or what's the most likely next response um and it kind of just remind me reminded me of like this generalized synchrony where the birds are singing the song and then the next segment of the song would be sort of it's kind of like a response almost to the other bird if there's like two birds singing um yeah I guess I was just like Curious in like how that and communication works like is this like the very most basic form of communicating in active inference like between two sorts of Agents uh like yeah yeah good questions Andrew oh yeah I just wanted to I uh previously skimmed a paper by friston and Par and they actually reference it here at the bottom of 165 um it they built another model that basically has two agents you know communicating with one another and with synchrony I I'd rather go with the bird song example I guess since it's more top of mine but um I it's just it's rather fascinating there's this um described like the birds are basically learning another's model parameters right such that they start making virtually the same inferences you know one is with the other um I think this is a wonderful kind of example too because I can imagine two musicians you know wanting to play the same song staying in time with one another kind of following the same script or sheet music um but yeah I'll I'll try to find that that paper just because I think it's if you're if you find interest in it it's you know it's more based on potentially what humans would be doing um it includes the entire model of like how do you you know how do you construct um in communication like what is a question um what are the different states uh involved with questions asking one another questions answering questions and and like so I'd love to check that out yeah yeah that's cool yeah you even said like scripts and uh Alber Ro n all with the strong and weak scripts so you have two musicians it could be they could literally have the same sheet music and that could be truly isomorphic sheet music like they're both playing the first trumpet part on this one song the same time it could also be a general synchrony which is to say not necessarily a lock step maybe it's a call and response so there's still a perfect Mutual information between the two sheet music readings that are ongoing even though they might be playing different instruments or they might be playing different parts so that's like what in that bird song example and subsequently they've referred to as like the singing from the same hym sheet and then kind of stepping out into the Improv is another step that I know some music and other researchers are are interested in because like with questions or other um turn-taking mechanisms because obviously it's different when there's epistemic transfer in communication of of whatever kind versus just the co- performance of something that's fixed ahead of time so that's that's one key piece and then um the uh the point about the large language models is a is a really good one so the the likeliest continuation of this trained model it's almost like the current token is y That's the observation and then we have like Q big foundation model Q now it's not necessarily a variational basian model but then there's going to be some variational free energy landscape and then when the temperature is low it picks simply the next likeliest outcome and then when the as the temperature increases you flatten that landscape so it starts sampling like more broadly to to the point of breaking out of kind of semantic ruts but decreasing potentially like the fluency or the coherence so that that's like we have the path of least action and then we have that temperature or shaky hand parameter because in language selection of topic and sentence and word and phon name and all of that is like this nested policy selection task and then so we have both the um path of least action that's kind of the ridge on the bottom of the valley and then we have this like sampling technique where you can sample along the strictest um likeliest path of continuation or you can sample more broadly like from that Valley so it's an example of how like statistics or otive inference approaches could be taken even if GPT 4 whatever doesn't need to be applying active inference from the inside but we could still talk about its expectations or preferences and how that conditions its actions and all these different aspects but it's map not territory so we'll be doing like cognitive mapping of gp4 not not reverse engineering or engineering from the inside so the bird song example is a singing from the same hym Sheet example where there's an expected and so it's an interesting and a good good uh continuation from the sensory attenuation they both have this an expectation in their gener each bird in in its generative model of how the song ised to go and then the question is are they going to be singing that song or not and so in this example it's like well like someone should be singing it this way and then it finds itself singing and then as it continues to sing takes its turn its um Precision drops we we could talk about why Precision might drop through time and then eventually that leads to a uh consideration hey maybe maybe I'm not the one continuing this song and then that's like the kind of turn-taking mechanic but it's not the only turn-taking mechanic that you could design you could use like a stop you know word like over or something like that or start words or all you know you can you can make any mechanism and then I think they show in that first paper with frisen frii or in some other papers like maybe what one have you mentioned about like the learning of the model and in that case there was like a younger bird like a more plastic bird with low precision and then over multiple rounds it leared how the song was supposed to be and then the first bird was taken away and then the new one was added in and they showed that basically you could do like this successive passaging of entrainments in the generalized synchrony setle so they mentioned like this leads to the Birds trying to like match their internal models um by trying to match how they continue singing um and I was I mean I definitely want to check out like how it was like all of this was like how fren and all of them connected this to communication um because like I guess I'm like wondering like how you communicate bits of information uh directly because like it makes sense you match the internal models and then you uh I guess like you take turns and you communicate the next section and then I guess what it informs the other bird just like from like a sort of based like information Theory type uh perspective it's like now the other bird knows that they're singing the same song um um I guess that is like some information that's being communicated and then um yeah it's like matching the same song and then I guess uh I guess that is kind of like an analog for language if you consider every idea or concept a song and you're trying to match with the same concept ccept or like some communication that you're trying to communicate I think there's some some I was just gonna say I think there's some some Theory somewhere I don't want to misquote someone I think it was from a neuroscientist uh um but uh the idea that whenever two people are communicating it seems to be the case that including using like neurophysiological measures like EEG or otherwise that there tends to be an increasing kind of synchrony between them um that includes in the in the kind of spectral Dimension like frequency um like if you did like a spectral decomposition of the EEG and look at the frequencies going on like their their frequencies tend to slowly begin to match one another um and furthermore if you kind of analyze the contents of the conversation which I'm not quite sure how you do this with high Precision mathematically but um the idea is like two people communicating with each other are slowly reaching some kind of convergence or consensus in what the conversation is about um about agreements upon the topic even if you disagree at some point you do converge in the idea of well now I know your stance and I also know my stance so at least that kind of Truth has been converged upon so I guess to Summit up like communication is some kind of General uh process of of convergence or syy of of information of brain frequencies um and I guess I guess here with um chapter eight uh synchron synchronization of chaos because I guess we haven't really gotten into that but like using the Lauren system as one potential like continuous model as well sharing all that that reminds me of like how communic a in uh like Wolf's ruad is sort of like between observers is kind of explained it was like uh between The Observers they have like shared like segments of the ruad um that they could both like kind of agree upon um which kind of reminds me of sort of like trying to match your internal model or you know like what you were saying like at least you converge that you both or or some understanding even if it's on an agreement it's like you're converging on how your stance is related to the other person's stance where like that relationship itself is kind of like a concept on its own um which is very interesting it's like we each we we're putting both of our cards on the table now we can see each other's cards so to speak like that information has not been shared um right even if the information that you brought to the table uh was different uh right now you've kind of come together here you have shared information and furthermore you are in the an instance of a particular conversation right so it's yeah yeah and like in information Theory like communication channels um it's kind of like um you're trying to it I mean a lot of this chapter reminded me a lot of communication channels from information Theory just like trying to know what the original input was that goes across a channel and then you get an output but the output might not be the same as the input and you have to see if it matches and if it's if there's noise then there might be some variation with the output so it's just kind of like trying to decrease that noise and then that's why you calculate the capacity of what's sort of the max rate of bits you could communicate across this channel without sort of um where you know for a fact that um you'll be able to map it directly to sort of the original input or like the cause so yeah a ton of great connections I think it's gonna be really exciting to see the quantum FP Chris Fields semantic information flow meet up with wolf from ruad I think that should be impending and but um yeah and then there's kind of like the the agent environment communication that's Alice and Bob and then sometimes we can kind of take the um Alice as agent environment as agent Bob as agent and then just sort of like summarize it with just a holographic screen with the classical information which is like saying that there's an inert uh intermediating entity so Quantum on either side of the boundary and classical or n dimensional on the boundary and then n plus dimensions on the in the quantum um and then there's uh simpler examples too like having a bunch of metronomes on a table so if the table is fixed and then there can't be like force transduction amongst the metronomes they will continue unabated with whatever population distribution is initiated and then if there is coupling nonzero coupling then like that's the kind of famous videos where they enter into lockstep they're simple or conservative active things so the general synchrony that they come to is like either all aligned or a two-phase but then I've also seen it where it's two-phase and then it kind of Falls a little bit off and then it actually slides all the way to back to lock step because that's a slightly more favorable synchrony but they're all exact versus one where say it's like 5050 or 991 with one going in the counter phas so that's kind of like the simplest generalized synchrony and then it's a huge open topic with semantics and Linguistics seems like there's a lot of information in the sequences of tokens how many bits how many knots relative to what but also it's not just the sequence of tokens we know that in our generation of language and our seemingly our perception that of course timing tone context all these other features come into play so then what is their Mutual information and there's settings where maybe just the tokens are necessary and sufficient there's another situation where the Tok there might not even be a token like a silent sign what does that say so then yeah these are great questions of communication I think we'll continue to see like in a lot of different fronts like ratcheting of the okay now sophisticated and the affective and and then they have these models of the nested look at this this is like figure 4.3 identical on top and then the bottom whereas in figure 4.3 in the textbook the bottom is the continuous time GM here we see the discrete time GM converted um into the for Factor graph and that supports the message passing more approximately but but we're dealing with this exact same discreete time model and then as we see with like the folk psychology models often the more cognitive or symbolic tasks are discreet and then the hybrid model is like where it hits the ground it's continuous like for example um the larynx um contraction or voice box or lip or tongue might be a continuous time continuous actuation model so you have some continuous um um voice apparatus with a propri reception and the actuation making phonemes but then once you're in the space of identifiable phonemes um or lexim then you can be at this level where they're now making discrete models at the symbolic so they've abstracted away from the phon they're not de with that here I'm saying that's that's kind of implicitly underneath it they're just saying that that they discreetly produce and recognize these words but then that's what's so nice about the composability of ACM okay now here they're taking it deep we see that four3 but now there's each time step has three nested so that's discreete with Indiscreet nesting and then you could also um just say instead of the observation being a discret token you could plug in a continuous model there and then you can evaluate you know empirically or what are the computational requirements or where is it where you getting more um value at and I think well the chaos thing that's explored a lot in the uh stochastic chaos and marov blankets paper I think 34 live stream 34 and then it's interesting how like even though a lot and I think Thomas par may have remarked on this in the book stream 2.1 like the earliest ACM models were the continuous time and then to accommodate classically symbolic cognitive processes and also explicit planning as inference and also to to integrate with a PDP well known in planning there was a movement towards the discret and then there's theost at all 2020 like a synthesis of active inference in discreet State spaces so not like it's like finished and people continue to develop new motifs and discrete compositions but kind of like the fundamentals of the discret were then focused on more and then potentially now with the path-based formulation of basian mechanics G Theory everything that Dalton and Maxwell at all are working on there's kind of like a a a a renewed emphasis and generalization on continuous structures like paths sheath bundles topos things that the book book doesn't necessarily go into but sets the stage for and then figure 8.6 kind of like um shows in the simplest possible example which is the same one that live stream 46 the ACM does not contradict folk psychology the simplest hybrid model where you have a discreet higher level symbolic um Dimension and then this kind of like sensory motor continuous time so this is like figure 4.3 get smooshed together and then yeah I mean just in terms of the location in the book second half is the more pragmatic uh model you know construction and implementation based Focus starting with chapter six with the recipe um and then just like the low and the high road were sort of this dialectic that that are unified or contained you know of or by active inference in seven and8 we have that distinction and then again with with this figure 8.6 by the end of chapter eight we're already landed with the synthesis of seven and eight whereas two and three they really are like different branches and they meet in the middle in seven and eight it's more like I mean they're just tucked together at the end of eight and then chapter nine introduces okay well what if instead of of specifying a GM hasht chapter 6 and then generating synthetic data what if we already had collected data of a certain structure we knew that we were going to collect data of a certain structure how could we then prepare a GM that would recognize that data allow us to do statistics on it whether we do discret continuous or Hybrid models chapter nine and then the end of the book so it's just like so funny how second half starts fast describes a few key here's the vertebrates here's the invertebrates or or whatever uh here's the kingdom of the fungi here's the other Kingdom then chapter nine gets more empirically oriented and chapter 10 simply reviews and contextualizes active inference so it's a quick ride it is very fast yeah let's just see what the other questions were how is five figure 52 related to figure 84 52 hierarchical predictive processing 84 luren synchronization we should we should procedurally generate every single possible question how is everyone every every figure I mean um I mean for a for a brief second I thought maybe it would related it to 8 figure 8.1 which is a little more I don't know um not quite message passing but at least it's you know we're still in the realm of looking at error terms basically these these basic equations with Lauren's to go with Lauren's systems though and relating that to message passing um I'm not sure I'm not so sure there's no clear there's no yeah no where my brain quickly goes well that's the attenuated space where any account could be possible or more accounts could be possible like again to this kind of question prompt question answer or even hegelian problem reaction solution structure it's interesting like which questions just have an answer which ones are like the Grand Canyon where the continuation there should only be like one specific continuation like what's the relationship between policy and affordance there may be a formal answer not even to say that that would be the only answer still but there might be a Sumo expressable necessary sufficient answer which is like pretty good but then for for for any question but especially some of them these questions are basically just evocative which is totally fine I think fun we're only going to experience even just with a 30 or 50 or how many figures there are I mean that would be like 200 questions MH so so just like then so that's the epistemic foraging which question should we ask prompt engineering conversation Theory all these different things like folding back into our the way that we're learning active like okay now that we've looked at what are the causes and consequences of sensory attenuation now what are we going to pay more or less attention to or now that we've talked about communication along these lines then how do we do what lot of fun options there here's the improvisation question yeah I'll just move it to this any any uh last thoughts um yeah so in chapter nine is it more about like using active inference on sorts of like if you wanted to use it on like a data set to detect something or you know things like that um yeah basically it points that way it is not the guide book or the Google collab notebook or whatever that just oh yeah load in your CSV and then this is just going to analyze the data for you so it's not to that point but what I think nine does that's important first 9.1 is a schema of the behavioral scientific setting where you have a behavioral science laboratory making a cognitive model of the rat or the ant or whatever which might be a map back of the environment but it frames it in a way that we can handle with ACM and then analogous to how chapter six gave a recipe for making the GM figure 92 gives us the road map from going from the data plus the GM to basically the kind of figure that would be in a published paper like we proposed um a personality variable or a psychometric variable of this kind continuous or discreet or whatever it was we had the generative model we had these data and likelihood function we inverted the model in light of our prior beliefs which might be uniform across groups or how whatever whatever it was in the setting and we think and this is kind of like Ryan Smith's Precision Psychiatry work you'll see a ton of things like this it's like and people who had this or this categorization or this or that continuous variable here was the correlation or the regression between our inference about this parameter and whether or not they were diagnosed this way and here's the T test and it um looks like something that you'd find in a paper in other words supporting an empirical claim being part of a fabric of evidence about an empirical system rather than just having a hypothetical GM and then speculating like it could be this way or it could be that wayc I mean and until you parameterize it and especially fit of data because it usually could so kind of like a six and little bit of a Jimmy Hendrick thing okay all right thank you fellas see you um next time see you next week bye bye thank you byebye